Bo Johnson — Curriculum Vitae

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Education

Indiana University (IU) **Bloomington, IN**
Ph.D. Candidate High Energy Physics *Aug. 2020–Present*
Graduate courses in classical mechanics, quantum mechanics, electricity & magnetism, statistical mechanics, quantum field theory, subatomic physics, digital electronics, nuclear astrophysics

Indiana University (IU) **Bloomington, IN**
MS Physics *April 2022*
Degree conferred for passing qualification exam

Utah State University (USU) **Logan, UT**
B.S. Physics, 4.0/4.0 *April 2020*
Minors in Mathematics and Computer Science

Brigham Young University (BYU) **Provo, UT**
B.S. Finance *April 2015*
Courses in financial derivatives; investments; money, banking, and financial systems; real estate; and investment management

Experience

Physics

Indiana University **Bloomington, IN**
Research Assistant (Dr. Rex Tayloe) *April 2021–Present*

- Member of the COHERENT Collaboration measuring CEvNS cross section on liquid argon
- Built and installed xenon gas system for injection to CENNS-10 for xenon-doping study
- Characterized PMTs for cyrogenic use in upcoming COH-Ar-750 experiment, including waveform analysis, fitting, and algorithm development

IU Physics Lab Instructor/Course Grader **Bloomington, IN**
Indiana University Dept. of Physics *Aug. 2020–April 2022*
Taught introductory physics labs in mechanics and electricity & magnetism to 120+ students, developing scientific and experimental understanding along with scientific communication for a wide variety of students. Graded weekly lab reports along with homework for the upper-division undergraduate quantum mechanics course.

Los Alamos National Laboratory **Los Alamos, NM**
Intern, Department of Energy Science Undergraduate Learning Initiative *May 2019–Aug. 2019*
Worked in the Neutron Group, studying how scintillator dead layers affect light output efficiency for use in experiment to measure the neutron lifetime.

USU Research Assistant: Working on non-linear dynamics problem involving magnetic spheres. Research includes numerical analysis in finding periodic modes of bouncing orbits and characterizing the stability of orbits.

USU Math Tutor Lab Instructor: Taught at Utah State's tutoring center for anyone at the university or community to get short tutoring help with courses from pre-algebra to differential equations.

Private Tutor: Instructed in math and physics courses to veterans returning to school.

USU Undergraduate Teaching Fellow: Resource for students taking Unveiling the Anthropocene course, a general education science course for art students learning about basic scientific principles in the context of environmental studies, creating homework assignments and grading homework.

USU Physics Recitation Leader: Reviewed concepts and homework for introductory physics courses PHYS 2210 for two semesters.

USU Physics Lab Instructor: Taught introductory physics laboratory for over 25 students.

Finance

Greener Equity **American Fork, UT**
Financial Analyst *2015–2016*
Involved in 350+ valuation projects in multiple analyst roles, valuing pre-revenue companies to $100+ million revenue companies. Interacted with executive managers of companies discussing valuation techniques used with their valuations. Trained new analysts, interns, and team members on valuation models, reports, and techniques.

The Boeing Company **Bellevue, WA**
Financial Analyst Intern *May 2014–Aug. 2014*

Increased reporting accuracy of capital expenditures for internal information technology customers by creating documents summarizing transaction accounts. Collaborated on summer project improving the training and reporting environment for 150+ IT financial analysts.

Publications & Presentations

- **Candidacy Seminar** (Talk). A ton-scale liquid argon detector for the COHERENT Collaboration: COH-Ar-750
- **2022 APS DNP New Orleans**. (Contributed Talk) "Status of CEvNS on LAr from the COHERENT Collaboration"
- **IU Physics/CEEM Summer 2022 Student Seminar.** (Talk) "Intro to Particle Detectors: Scintillators and Photodetectors."
- The COHERENT Collaboration *et al*, "A COHERENT constraint on leptophobic dark matter using CsI data." 24 May 2022. https://doi.org/10.48550/arXiv.2205.12414
- The COHERENT Collaboration *et al*, "The COHERENT Experimental Program." Snowmass contribution. 10 April 2022. https://doi.org/10.48550/arXiv.2204.04575
- The COHERENT Collaboration *et al*, "Monitoring the SNS basement neutron background with the MARS detector." Journal of Instrumentation 17 P03021. https://doi.org/10.1088/1748-0221/17/03/p03021
- Boyd F. Edwards, **Bo A. Johnson**, and John M. Edwards. "Periodic Bouncing Modes for Two Uniformly Magnetized Spheres. I. Trajectories." Chaos: An Interdisciplinary Journal of Nonlinear Science 30, no. 1 (January 2020): 013146. https://doi.org/10.1063/1.5125924.
- Boyd F. Edwards, **Bo A. Johnson**, and John M. Edwards. "Periodic Bouncing Modes for Two Uniformly Magnetized Spheres. II. Scaling." Chaos: An Interdisciplinary Journal of Nonlinear Science 30, no. 1 (January 2020): 013131. https://doi.org/10.1063/1.5125925.
- **2018 Annual Meeting of the APS Four Corners Section at the University of Utah.** (Poster). **Johnson, B.**, & Edwards, B. Periodic Solutions to Two Magnetic Sphere Interactions. Bulletin of the American Physical Society, (63)16, G01.00052
- **Utah State University Physics Colloquium Presenter.** September 2018. Interactions between Space Weather and Atmospheres
- **2017 Annual Meeting of the APS Four Corners Section at the University of Colorado.** (Poster). **Johnson, B.**, & Sojka, J.J. Comparison of Neutral Planetary Atmospheres with Space Weather Events. Bulletin of the American Physical Society, 62(17), G01.00005.
- **Utah State University Physics Colloquium Presenter.** October 2017. Comparison of Neutral Planetary Atmospheres with Space Weather Events.

Honors and Awards

DOE SCGSR Fellowship 2023: This fellowship provides graduate students the opportunity to pursue thesis research at a national laboratory. My proposal was titled COH-Ar-750 Development and Deployment on behalf of the COHERENT Collaboration.

Dean's Scholar: Awarded to students who have achieved a perfect 4.0 GPA throughout their undergraduate studies within the College of Science at USU.

Outstanding Graduating Senior: Awarded to a graduating physics senior for high academic achievement and professional promise.

Hendricks Scholarship 2019: Awarded based on academic merit, professional potential, and personal integrity.

Howard L. Blood Fellowship 2018: Summer research fellowship.

O. Harry Ottesen Award 2018: Given to physics majors achieving the highest score in general physics.

Sigma Pi Sigma Physics Honor Society Member: Extended to those who demonstrate academic excellence in physics.

Howard L. Blood Fellowship 2017: Summer research fellowship.

Selected skills

Programming proficiency: C/C++, Python, Julia, git, Mathematica

Software: Linux, CERN ROOT, Geant4, LaTeX, LibreOffice, Microsoft Office

Computer experience: Numerical methods including ODE integration, nonlinear equation solving, and physical model simulations. Creating GUIs, data structures, and algorithms.

Radiation detectors: Cryogenics, DAQ systems, photodetectors

Organization Memberships

- **2018–** $\Sigma\Pi\Sigma$ Physics Honor Society
- **2017–** Society of Physics Students